
Note: If you are using Python, it is recommended to submit all your programs in a Jupyter notebook. If you are using a different programming language, you can have multiple programs for each of the questions below. Please employ clean code writing practices so that your code is comprehensible to others.

There should be two components to your submission: a written component with all the solutions / conclusions of the various problems, and a folder with the relevant code. Note that some problems simply require you to derive something rather than write a program.

Choose a decent number of bins for the histograms that you have to plot in the problems below. Too few or too many bins will make the histograms either show very little detail or too noisy. I find it reasonable to have 50 bins, but you should tune this according to the samples you get.

1 Sampling continuous random variables

In the previous assignment, we sampled a discrete random variable (the coin toss random variable) which has value H (equivalently $+1$) with probability p and value T (equivalently -1) with probability q with $p + q = 1$. The method followed is the tower sampling method where we randomly generate a number between 0 and 1 with uniform probability, and call the outcome H if the number is between 0 and p , and T if the number is between p and 1.

In this assignment, we discuss a similar method to sample continuous random variables. The cumulative distribution function (cdf) $F_{\mathbb{X}}(x)$ of $\mathbb{X}$ gives the probability $\mathbb{P}\{\mathbb{X} \leq x\}$ for the random variable $\mathbb{X}$. We work with random variables whose cdf's are continuous, and in most cases, differentiable as well¹. In this case, the probability $\mathbb{P}\{a \leq \mathbb{X} \leq b\}$ that $\mathbb{X}$ lies in an interval $[a, b]$ is given by $F_{\mathbb{X}}(b) - F_{\mathbb{X}}(a)$. The probability that $\mathbb{X}$ lies in the infinitesimal interval $[x, x + dx]$ is then $F_{\mathbb{X}}(x + dx) - F_{\mathbb{X}}(x)$.

The cdf $F_{\mathbb{X}}(x)$ is a monotonically increasing function over its domain $(-\infty, \infty)$, and has range $[0, 1]$. This means that the inverse function exists for $F_{\mathbb{X}}$, with domain $[0, 1]$ and range $(-\infty, \infty)$ and is also continuous and monotonically increasing. This inverse function is usually denoted $F_{\mathbb{X}}^{-1}(y)$ where $y \in [0, 1]$.

To obtain a sample of the random variable $\mathbb{X}$ that is distributed according to the cdf $F_{\mathbb{X}}(x)$, we first generate a random number y which is uniformly distributed in $[0, 1]$ (this is what

¹A general cdf is typically only right continuous with a left limit. That is, it is typically a function $F(x)$ such that $\lim_{x \rightarrow a^+} F(x) = F(a)$ and $\lim_{x \rightarrow a^-} F(x)$ exists but is not necessarily $F(a)$. Such functions are called càdlàg functions or RCLL functions (Right-Continuous-Left-Limit). Such functions are also useful in describing paths of stochastic processes. We shall not be concerned with such subtleties in this assignment.

`random.random` does in Python). Then, we apply the inverse function $F_{\mathbb{X}}^{-1}$ to the randomly generated y and obtain the sampled value of $\mathbb{X}$ as $F_{\mathbb{X}}^{-1}(y)$. It is easy to show that if y is uniformly distributed, then $F_{\mathbb{X}}^{-1}(y)$ is distributed according to the cdf $F_{\mathbb{X}}(x)$.

Once we generate N samples from the uniform distribution on $[0, 1]$, we get the corresponding N samples $F_{\mathbb{X}}^{-1}(y)$. Binning these N samples with bin size Δx , it is easy to show that the number of counts in the bin $[x, x + \Delta x]$ is given by $F_{\mathbb{X}}(x + \Delta x) - F_{\mathbb{X}}(x)$.

When the cdf of a random variable is differentiable, one can write down a probability density function (pdf)

$$f_{\mathbb{X}}(x) = \frac{d}{dx} F_{\mathbb{X}}(x) . \quad (1)$$

This has the interpretation that the probability that $\mathbb{X}$ takes values in $[x, x + dx]$ is $F(x + dx) - F(x) = f_{\mathbb{X}}(x)dx$. We also have

$$F_{\mathbb{X}}(x) = \int_{-\infty}^x dz f_{\mathbb{X}}(z) . \quad (2)$$

The lower limit can be something else depending on the nature of the random variable $\mathbb{X}$. For instance, for a random variable that takes values in $[0, \infty)$, the lower limit is 0 rather than $-\infty$.

Recall that the number of counts in the histogram of N samples $F_{\mathbb{X}}^{-1}(y)$ in the bin $[x, x + \Delta x]$ will be $F(x + \Delta x) - F(x)$. When there is a pdf, this will be approximately equal to

$$F(x + \Delta x) - F(x) \approx f_{\mathbb{X}}(x)\Delta x , \quad (3)$$

and the approximation gets better and better as the bin width Δx becomes smaller and smaller.

Exercise 1a (4 points): This exercise has to be repeated for each of the following values of N :

1. $N = 100$

2. $N = 1000$

3. $N = 10000$

Generate N samples from the uniform distribution over $[0, 1]$. This is simply done by running a random number generator like `random.random` by passing `size=N`. To get a random number z uniformly distributed in the interval $[a, b]$, we transform the samples y from $[0, 1]$ using

$$z = a + (b - a)y . \quad (4)$$

Obtain N random numbers in the interval $[0, 2\pi]$. Plot the histogram of the N samples and check that the normalized histogram approximates the pdf of the uniform distribution

$$f(x) = \begin{cases} \frac{1}{2\pi} & 0 \leq x \leq 2\pi , \\ 0 & \text{otherwise.} \end{cases} \quad (5)$$

Exercise 1b (4 points): Consider a random variable $\mathbb{X}$ which takes values in $[0, \infty)$ and is exponentially distributed. That is, its cdf and pdf are

$$F_{\mathbb{X}}(x) = 1 - e^{-\lambda x} , \quad \text{and} \quad f_{\mathbb{X}}(x) dx = \lambda e^{-\lambda x} dx , \quad \text{for some } \lambda > 0. \quad (6)$$

Show that the solution to $F_{\mathbb{X}}(x) = y$ for x given some y in $[0, 1]$ is

$$x = -\frac{1}{\lambda} \log(1 - y) . \quad (7)$$

This gives the inverse function of the cdf $F_{\mathbb{X}}(x)$:

$$F_{\mathbb{X}}^{-1}(y) = -\frac{1}{\lambda} \log(1 - y) . \quad (8)$$

The inverse function of the cdf is sometimes called the percent point function (ppf).

The following steps have to be done for each of the following values of N :

1. $N = 100$
2. $N = 1000$
3. $N = 10000$

and the following values of λ :

1. $\lambda = 1$
2. $\lambda = 0.1$
3. $\lambda = 2$

Generate N random numbers between $[0, 1]$ using your favorite random number generator. Apply the inverse function $F_{\mathbb{X}}^{-1}(y)$ and obtain the corresponding N samples of the exponential distribution. Plot a histogram of the N samples and check that the normalized histogram agrees roughly with the pdf $f_{\mathbb{X}}(x) = \lambda e^{-\lambda x}$, $0 \leq x < \infty$.

What happens to the agreement of the histogram with the pdf as N becomes large? ■

We will now describe how to sample a random variable $\mathbb{X}$ with a normal distribution. The pdf of the standard normal distribution is

$$\varphi_{\mathbb{X}}(x) = \frac{1}{\sqrt{2\pi}} \exp(-x^2/2) , \quad (9)$$

and the cdf is the integral

$$F_{\mathbb{X}}(x) = \int_{-\infty}^x dz \varphi_{\mathbb{X}}(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x dz e^{-z^2/2} . \quad (10)$$

Unlike the exponential distribution, there is no explicit expression for the cdf, and finding the expression for $F_{\mathbb{X}}^{-1}$ is not easy. Instead, we employ a useful trick that allows us to exactly generate the samples without having to calculate the inverse cdf.

Consider two independent random variables $\mathbb{X}$ and $\mathbb{Y}$ both identically distributed with the normal distribution. The joint pdf is then the product of the individual pdfs:

$$f_{\mathbb{X},\mathbb{Y}}(x, y)dx dy = f_{\mathbb{X}}(x)dx f_{\mathbb{Y}}(y)dy = \frac{1}{2\pi}e^{-(x^2+y^2)/2}dx dy . \quad (11)$$

Let us introduce new random variables $(\mathbb{R}, \Theta)$ with

$$\mathbb{X} = \sqrt{2\mathbb{R}} \cos \Theta , \quad \mathbb{Y} = \sqrt{2\mathbb{R}} \sin \Theta , \quad \mathbb{R} = \frac{1}{2}(\mathbb{X}^2 + \mathbb{Y}^2) , \quad \Theta = \tan^{-1} \frac{\mathbb{Y}}{\mathbb{X}} . \quad (12)$$

The random variable $\mathbb{R}$ has range $[0, \infty)$ and Θ has the range $[0, 2\pi]$. We obtain the pdfs $f_{\mathbb{R}}(\rho)$ and $f_{\Theta}(\theta)$ next.

Exercise 1c (2 points): Show that the distribution $f_{\mathbb{X},\mathbb{Y}}(x, y)dx dy$ is the product of a exponential distribution for $\mathbb{R}$ and a uniform distribution for Θ over $[0, 2\pi]$:

$$f_{\mathbb{R}}(\rho)d\rho = e^{-\rho}d\rho , \quad f_{\Theta}(\theta)d\theta = \frac{1}{2\pi}d\theta . \quad (13)$$

You will need the change of variables formula $dx dy = d\rho d\theta$ (no need to derive this). ■

Now, it is clear how to generate samples of the normal distribution. We first sample separately from the exponential distribution for $\mathbb{R}$ and from the uniform distribution for Θ using the results of Exercise 1b and 1a. After that, we transform back to $\mathbb{X}$ and $\mathbb{Y}$ using the formulas (12).

Exercise 1d (4 points): This exercise has to be repeated for the following values of N :

1. $N = 100$
2. $N = 1000$
3. $N = 10000$

Use the N samples of the uniform distribution on $[0, 2\pi]$ in Exercise 1a, the N samples of the exponential distribution with $\lambda = 1$ that you obtained in Exercise 1b. Label these samples by θ and ρ respectively. Generate N samples of the pair of random variables $(\mathbb{X}, \mathbb{Y})$ with

$$\mathbb{X} = \sqrt{2\mathbb{R}} \cos \Theta , \quad \mathbb{Y} = \sqrt{2\mathbb{R}} \sin \Theta . \quad (14)$$

Each of these should give samples from the independently distributed normal random variables $\mathbb{X}$ and $\mathbb{Y}$. Verify this claim by plotting a histogram of the N samples for each of the random variables $\mathbb{X}$ and $\mathbb{Y}$ and checking that the normalized histograms agree roughly with the standard normal distribution (9). ■

Exercise 1e (2 points): In this exercise, we will generate samples for a pair of random variables $\mathbb{X}_1, \mathbb{X}_2$ which are normally distributed with means μ_1, μ_2 and variances σ_1^2 and σ_2^2 respectively. The pdf for a normal distribution with mean μ and σ^2 is

$$\frac{1}{\sqrt{2\pi\sigma^2}} \exp \left(-\frac{(x - \mu)^2}{2\sigma^2} \right) . \quad (15)$$

These can be generated by the modifying the transformations (14) as

$$\mathbb{X}_1 = \mu_1 + \sigma_1 \sqrt{2\mathbb{R}} \cos \Theta, \quad \mathbb{X}_2 = \mu_2 + \sigma_2 \sqrt{2\mathbb{R}} \cos \Theta. \quad (16)$$

Choose the values $\mu_1 = 1$, $\mu_2 = -1$, $\sigma_1 = 2$, $\sigma_2 = 0.5$. Generate $N = 10000$ samples of the pair of random variables $\mathbb{X}_1$, $\mathbb{X}_2$ using the $\mathbb{R}$, Θ samples from Exercise 1d and using the transformations (16). Plot the normalized histograms of the pair of random variables and check that they agree with the pdfs of normal distributions with the appropriate means and variances. ■

We have generated a pair of normally distributed random variables that are independent. Next, we describe how to generate two normally distributed random variables which have a non-zero correlation between them.

Exercise 1f (4 points): Starting with two independent, normally distributed random variables $\mathbb{X}$, $\mathbb{Y}$, with means μ_1 , μ_2 and equal variances σ^2 respectively, form the new random variables $\mathbb{X}'$, $\mathbb{Y}'$ with

$$\begin{pmatrix} \mathbb{X}' \\ \mathbb{Y}' \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} \mathbb{X} \\ \mathbb{Y} \end{pmatrix}, \quad (17)$$

where a , b , c , d are real numbers such that the determinant $ad - bc \neq 0$. In other words, we have

$$\mathbb{X}' = a\mathbb{X} + b\mathbb{Y}, \quad \mathbb{Y}' = c\mathbb{X} + d\mathbb{Y}. \quad (18)$$

Calculate the expectation values μ'_1 , μ'_2 of $\mathbb{X}'$ and $\mathbb{Y}'$ respectively, in terms of the means and variances of $\mathbb{X}$ and $\mathbb{Y}$. Then, calculate the covariance matrix

$$\begin{pmatrix} \mathbb{E}[(\mathbb{X}' - \mu'_1)^2] & \mathbb{E}[(\mathbb{X}' - \mu'_1)(\mathbb{Y}' - \mu'_2)] \\ \mathbb{E}[(\mathbb{Y}' - \mu'_2)(\mathbb{X}' - \mu'_1)] & \mathbb{E}[(\mathbb{Y}' - \mu'_2)^2] \end{pmatrix}, \quad (19)$$

by using the relations (18) and that the $\mathbb{X}$, $\mathbb{Y}$ are independent normally distributed random variables with means μ_1 , μ_2 and equal variances σ^2 respectively.

What is the Pearson correlation coefficient $\rho(\mathbb{X}', \mathbb{Y}')$ in this case? Recall that it is defined as

$$\rho(\mathbb{X}', \mathbb{Y}') = \frac{\text{Cov}(\mathbb{X}', \mathbb{Y}')}{\sqrt{\text{Var}(\mathbb{X}')\text{Var}(\mathbb{Y}')}}. \quad (20)$$

Exercise 1g (2 points): Generate $N = 10000$ samples of two independent random variables distributed normally with mean 0 and variance 1 (you can use the samples from Exercise 1d). Form the new random variables $\mathbb{X}'$, $\mathbb{Y}'$ as in Exercise 1f with the matrix

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}, \quad (21)$$

and obtain the corresponding N samples of $\mathbb{X}'$, $\mathbb{Y}'$ from those of $\mathbb{X}$, $\mathbb{Y}$. Plot the N samples of $\mathbb{X}'$, $\mathbb{Y}'$ against each other in a scatter plot. Do you see a correlation between the two random variables? What is the Pearson correlation coefficient for the above choice of a , b , c , d ? ■

2 Brownian motion

In this part of the assignment, we simulate Brownian motion using our generator of normal random variable samples from the previous section. Since we generate the normal random variables in pairs, we discard one of the pair of values to get a single normal random variable.

Alternatively, for this section of the assignment, you can use a ready-made random sample generator for the normal distribution (for example, `scipy.stats.norm.rvs()` in Python).

Recall that Brownian motion or the Wiener process $\mathbb{W}(t)$, $0 \leq t \leq T$, is a stochastic process which is defined to satisfy the requirements

1. $\mathbb{W}(t)$ is a continuous function of t and satisfies $\mathbb{W}(0) = 0$,
2. $\mathbb{W}(t)$ for each t is normally distributed with mean 0 and variance t ,
3. Choose time instants $0 = t_0 < t_1 < t_2 < \dots < t_n < T$. The non-overlapping increments $\mathbb{W}(t_1) - \mathbb{W}(t_0)$, $\mathbb{W}(t_2) - \mathbb{W}(t_1)$, $\dots$, $\mathbb{W}(t_n - t_{n-1})$ are independent random variables, and $\mathbb{W}(t_{i+1}) - \mathbb{W}(t_i)$ is normally distributed with mean 0 and variance $t_{i+1} - t_i$.
4. $\mathbb{W}(t)$ is adapted to a filtration $\{\mathcal{F}(t)\}_{0 \leq t \leq T}$ with the increment $\mathbb{W}(t) - \mathbb{W}(s)$ independent of $\mathcal{F}(s)$ for $s < t$.

These properties directly suggest a way to simulate the Wiener process. First, we partition the time interval $[0, T]$ into N equal intervals by choosing equally spaced time instants $0 < t_1 < t_2 < \dots < t_{N-1} < t_N = T$ where $t_0 = 0$ is the initial time and $t_N = T$ is the final time. The spacing or ‘step’ between two consecutive time instants is

$$\varepsilon = \frac{T - 0}{N} . \quad (22)$$

Typically, N is very large compared to T and consequently, the step size ε is a very small number. To generate a sample of the Wiener process over $[0, T]$, we have to

1. start at $\mathbb{W}(0) = 0$ at $t = 0$,
2. at $t = t_1$, generate a sample from the normal distribution with mean 0 and variance $t_1 - 0$, and add the sample to $\mathbb{W}(0)$ to get $\mathbb{W}(t_1)$,
3. at $t = t_2$, generate a new sample from the normal distribution with mean 0 and variance $t_2 - t_1$, and add the new sample to $\mathbb{W}(t_1)$ to get $\mathbb{W}(t_2)$,

4. and so on, till we reach $t = t_{N-1}$, where we generate the final new sample from the normal distribution with mean 0 and variance $t_N - t_{N-1} = T - t_{N-1}$, and add this new sample to $\mathbb{W}(t_{N-1})$ to get $\mathbb{W}(T)$.

In other words, at each time instant $t = t_i$, we generate a new sample from the normal distribution with mean 0 and variance $t_{i+1} - t_i$, and add the sample to $\mathbb{W}(t_i)$ to get $\mathbb{W}(t_{i+1})$. We repeat this procedure for $i = 0, \dots, N - 1$.

One way to speed up this process is to realize that a normal random variable $\mathbb{X}$ with mean 0 and variance σ^2 is related to a standard normal random variable $\hat{\mathbb{X}}$ with mean 0 and variance 1 as

$$\mathbb{X} = \sigma \hat{\mathbb{X}} . \quad (23)$$

Recall that the variance of each of the increments $\mathbb{W}(t_{i+1}) - \mathbb{W}(t_i)$ is the time step $t_{i+1} - t_i = \varepsilon$. We can generate the samples of the increments at each stage of the Wiener process simply by first generating N samples of a standard normal random variable and multiply each of those samples by $\sqrt{\varepsilon}$.

Thus, the final algorithm is as follows.

1. Fix a time step ε by choosing an appropriate number of partitions N .
2. Sample the standard normal distribution N times, s_i , $i = 0, \dots, N - 1$, and multiply each of the samples by $\sqrt{\varepsilon}$. These give the successive increments $\mathbb{W}(t_{i+1}) - \mathbb{W}(t_i) = s_i \sqrt{\varepsilon}$.
3. Starting from 0, successively add the sampled increments to get the values of the Wiener process $\mathbb{W}(t_0), \mathbb{W}(t_1), \dots, \mathbb{W}(t_{N-1}), \mathbb{W}(T)$. (You can use `numpy.cumsum` if you are coding in Python.)

Note: This simpler algorithm cannot be used if the time intervals are not all equally spaced. In that case, one has to go back to the original method discussed above.

Exercise 2a (4 points): Simulate the Wiener process for $T = 1$, by choosing the number of partitions of $[0, T]$ to be $N = 1000$. Repeat this simulation $R = 10$ times. During each of these $R = 10$ times, you will get an realization of the random Wiener process.

Plot the $R = 10$ different realizations of the Wiener process in the same plot. You may want to choose different colours for each of the 10 plots to distinguish between the different realizations. Try to make the plot as visually appealing as possible. ■

Exercise 2b (6 points): This exercise has to be done for the following values of R :

1. $R = 100$
2. $R = 1000$
3. $R = 10000$

Generate the Wiener process for $[0, 1]$ with $N = 10000$ time steps (this means that $\varepsilon = 1/N = 10^{-4}$). Repeat the simulation R times.

Keep track of the values of the Wiener process for the following five time instants:

$$t = 0.2, \quad t = 0.4, \quad t = 0.6, \quad t = 0.8, \quad t = 1.0. \quad (24)$$

For each time instant you should have R values of the Wiener process that you obtain from the R trials. Separately for each of the time instants t given above, bin these R values and obtain a histogram. Verify that the normalized histogram follows the normal distribution with mean 0 and variance t by superimposing the theoretical pdf curve of the normal distribution with mean 0 and variance t .

At the end of this exercise, you should have three sets (for each of the three values of R) of five histograms for each of the five time instants above. What happens to the agreement of the histograms with the appropriate normal distributions as R increases? ■

Exercise 2c (4 points): In this exercise, we will compute the quadratic variation of the Wiener process. Recall that the quadratic variation for $\mathbb{W}(t)$ is defined as follows. As usual, we partition the interval $[0, T]$ into N pieces with the intermediate time instants $0 = t_0 < t_1 < t_2 < \dots < t_{N-1} < t_N = T$. Let us space these equally as well. Then, the quadratic variation is

$$[\mathbb{W}, \mathbb{W}](T) = \lim_{N \rightarrow \infty} \sum_{i=1}^N (\mathbb{W}(t_i) - \mathbb{W}(t_{i-1}))^2. \quad (25)$$

Of course, in practice one can never attain the limit $N \rightarrow \infty$. The best we can do is to choose a sufficiently large value of N and compute the right hand side.

Choose $N = 10000$ and simulate the Wiener process on $[0, 1]$. Repeat $R = 100$ times. Calculate the quadratic variation $[\mathbb{W}, \mathbb{W}](T)$ using the above formula for each of the 100 realizations of the Wiener process. Note that since the quadratic variation involves only the increments of the Wiener process rather than individual values at the time instants, you can compute the quadratic variation directly after you generate the N increments in Step 2 of the final algorithm.

Study the $R = 100$ values for the quadratic variation. Bin them and obtain a histogram. What is the distribution of the quadratic variation centred on? Repeat the above exercise for $R = 1000$ and comment on any difference in the histogram of the $R = 1000$ quadratic variation values. ■

The above exercise was to illustrate the fact that the statement $[\mathbb{W}, \mathbb{W}](T) = T$ holds only *almost surely*. In particular, in any finite number of simulations of the Wiener process, there will always be a non-zero spread about the value T .

Exercise 2d (4 points): In this exercise, we look at the maximum-to-date random variable $\mathbb{M}(t)$ associated to the Wiener process. This will be useful in simulating barrier options later.

The maximum-to-date or running maximum process is defined as

$$\mathbb{M}(t) = \max_{0 \leq s \leq t} W(s) . \quad (26)$$

That is, it is the maximum of the values of the Wiener process from time 0 to time t .

Take the $R = 100$ realizations of the $N = 10000$ Wiener process on $[0, 1]$ in Exercise 2c and obtain the corresponding maximum-to-date process for each of the 100 realizations. Choose any 5 of the 100 realizations and plot both the Wiener process and maximum-to-date process in the same plot. You should have 5 plots at the end of this exercise. ■

The paths of the maximum-to-date process are either constant or increasing over the time interval of the Wiener process. It is constant if the Wiener process values does not rise above the current value of the maximum process, and increases when the Wiener process crosses the current maximum value.

As you have seen in Exercise 2d, the paths of the maximum-to-date process $\mathbb{M}(t)$ are quite intricate even though they are only constant or increasing. In particular, it can be shown that, almost surely, there is **no interval** in time where the maximum-to-date function is increasing. In other words, even though there are many points where the maximum-to-date value increases (where the Wiener process crosses the current maximum-to-date value), these points do not occupy any interval of finite length. The correct mathematical notion to describe a set of such points is that of a *Cantor set*. A Cantor set has an infinite number of points, but the set itself has length zero since the points in the Cantor set are not populous enough to fill up an interval of any non-zero length. This is an example of a *fractal*.

Exercise 2e (Optional): Look at the $R = 100$ realizations of the maximum-to-date process in the time interval $[0, 1]$ that you obtained in Exercise 2d. For each of these 100 realizations, compute the sum of the lengths of the intervals where the path $\mathbb{M}(t)$ is constant (these are the intervals where the Wiener process does not cross the current maximum value). Plot a histogram of these 100 values and show that they all cluster around the value 1 (the length of the time interval for the Wiener process). Redo the exercise for $N = 10^5$ and $N = 10^6$ time instants in the interval $[0, 1]$, with $R = 100$ realizations in each case. Show that the clustering becomes stronger as N increases. ■